

CODEMIX SKILL

The safety + decision layer between AI agents and their actions

*“Order cancel **mat** karo, bas address change karna hai.”*



We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

1

Initiative Overview

Initiative: Codemix Skill · Owner: Team Ramanathan & Sadhana · a plug-in that makes tool-using AI agents safe

The gap

Voice agents can understand a request and immediately call a refund, cancellation or replacement API. A small missed negation can turn the right conversation into the wrong transaction.

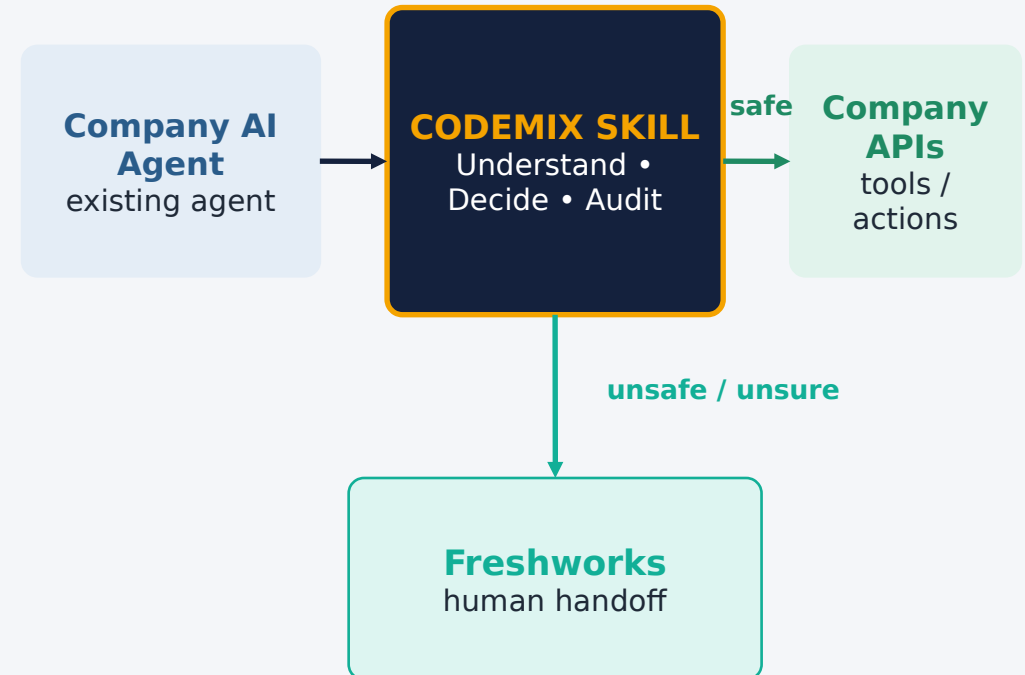
Our product

Codemix Skill sits between the AI agent and its API/tool calls. It validates intent, negation, confidence, sentiment and SOP rules before execution.

Freshworks extension

Unsafe, uncertain or emotionally escalated calls become Freshdesk tickets with transcript, decision context and action details for a human agent.

WHERE IT SITS



The agent proposes an action. Codemix Skill decides whether it runs, needs confirmation, or goes to a human.

2

Problem Statement

In code-mixed voice, one short word can reverse the meaning of an API action

HINDI EXAMPLE

*“Order cancel **mat** karo, bas address change karna hai.”*

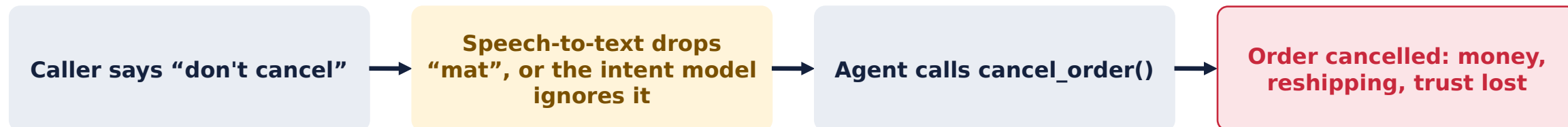
Correct intent: do NOT cancel → update address

TAMIL EXAMPLE

*“Refund **vendam**, replacement anuppunga.”*

Correct intent: do NOT refund → send replacement

HOW IT BREAKS TODAY



48%

of negated requests would still trigger a refund / cancel in our own Round-1 engine

Measured: 11 of 23 team-written test sentences (Hindi, Tamil, English, Telugu, Kannada, Bengali)

26%

of cash-on-delivery orders in India return to origin, so refund / cancel / return calls are high-volume

Source: Shipway ShipNotes report, FY25 (Jul 2025)

3

Why Now

AI agents have write access, so misunderstanding is no longer just a bad answer



AI agents can act

Tool calling and MCP let agents trigger refunds, cancellations, returns and other business actions directly.



Code-mixing is normal

Indian callers switch between languages inside a sentence. The risk is concentrated around small intent-flipping words.



Voice stack is ready

VoBiz provides phone access + DTMF, Sarvam handles Indian speech-to-text, Claude validates intent, and ElevenLabs returns the response.

27-70%

word error rate on Hinglish across speech-to-text models, on identical audio. Speech recognition alone can't yet be trusted with irreversible actions, so the gate is needed now. *Source: Deepgram, "Hinglish Voice AI: Why ASR Fails"*

Our timing The missing layer is not another voice bot. It is a safety and decision layer that controls when an existing AI agent is allowed to call an API.

4

Competitive Landscape

We focus specifically on the gap between language understanding and action execution

Do nothing

- ✓ Zero integration effort.
- ✗ Risky actions can execute on the agent's first interpretation.

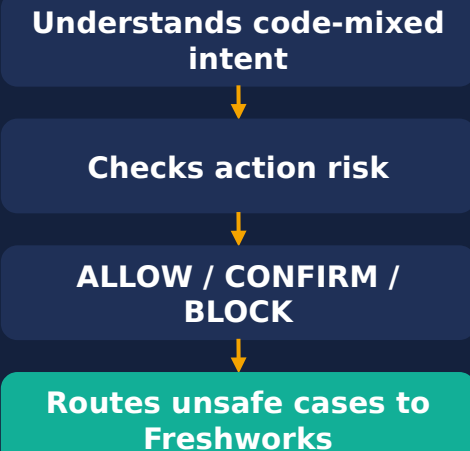
Confirm-everything IVR

- ✓ Safe confirmation.
- ✗ Adds friction to every action, even low-risk requests.

Generic LLM guardrails

- ✓ Useful for policy, toxicity and PII.
- ✗ Not designed as an action gate for code-mixed negation + speech confidence.

Codemix Skill



Friction on every action

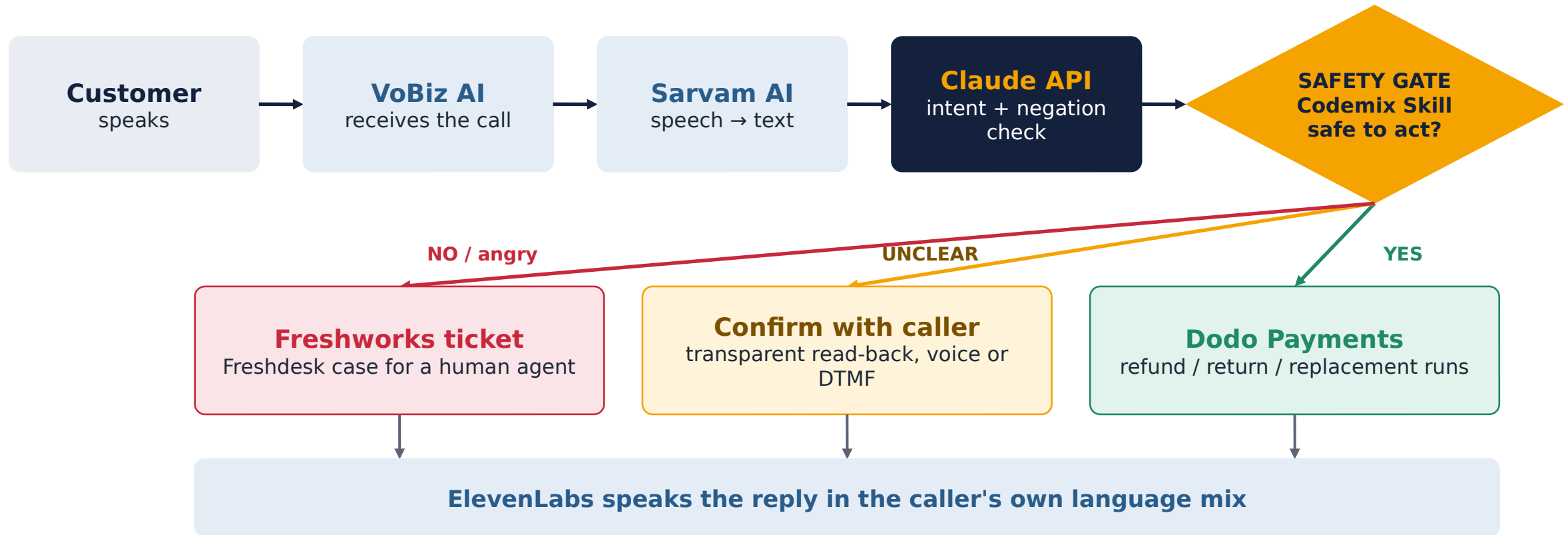
Risk-based: only risky actions get a check

Our edge A policy-aware action gate with human escalation and an auditable record of what the caller said, what the agent decided, and what API was called.

5

Solution Architecture

Codemix Skill is the control point between an existing AI agent and its tools



Our innovation We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

5

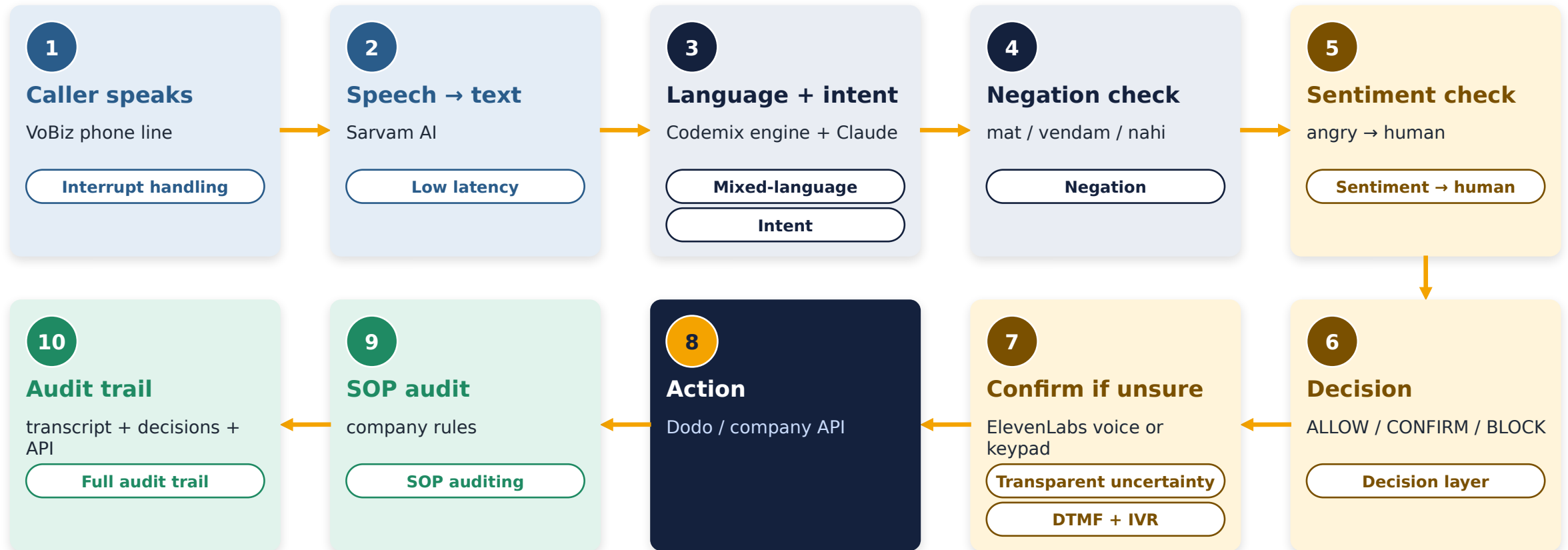
Sponsor & Integration Stack

Each sponsor/API has one clear job; Codemix Skill coordinates the handoff and controls action execution

CHANNEL	VoBiz AI	PHONE + DTMF	The number customers call. Keypad (DTMF) gives a confirmation that needs no speech recognition.
LISTEN	Sarvam AI	SPEECH → TEXT	Transcribes code-mixed Hindi + English and Tamil + English, so the negation reaches the intent check.
VALIDATE	Claude API	INTENT + SAFETY	Before any sensitive action: finds the real intent. “Refund vendam” = NO refund, YES replacement.
CONTROL	Codemix Skill	ORCHESTRATION + SAFETY	The safety gate between what the voice agent hears and what it is allowed to do.
ACT	Dodo Payments	ACTION API	Runs refunds, returns and replacements, but only after the safety gate says YES.
ESCALATE	Freshworks / Freshdesk	HUMAN WORKFLOW	If it can't be resolved safely, a Freshdesk ticket hands the case to a human agent.
RESPOND	ElevenLabs	VOICE OUTPUT	Speaks the reply in the caller's own language mix and reads back what it understood.

5 One Call, End to End — All 11 Features

Every feature appears only where it adds value; routine calls take the fast path



Sentiment: Claude scores caller sentiment every turn; two negative turns or an explicit request for a human → Freshdesk ticket.

Low latency: fast offline checks first (1.87 ms measured); Claude runs only before sensitive actions. White tags = the 11 features.

5

The Safety Decision & Transparent Confirmation

The agent never silently guesses on an irreversible action



TRANSPARENT CONFIRMATION IN PRACTICE

CALLER

"Refund... vendam... maybe replacement."

AGENT • ElevenLabs, caller's mix

"I'm not fully sure whether you want a refund or a replacement. Press 1 for replacement, 2 for a human agent."

CALLER • keypad via VoBiz

presses 1 → no speech-to-text ambiguity

Interrupts: VoBiz voice-activity detection stops ElevenLabs playback the moment the caller speaks; the new words become the latest input.

BLOCK / ROUTE

Negation conflicts with the action, or confirmation fails → stop API + Freshdesk.

EXAMPLE OUTCOME

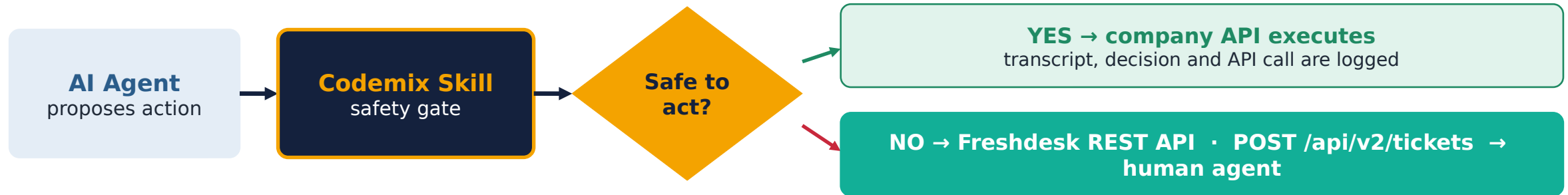
"Refund vendam, replacement anuppunga."

BLOCK refund → confirm replacement → if unclear, Freshdesk ticket

5

Freshworks Integration — Human-in-the-Loop

Freshdesk takes the handoff; also works with Freshcaller as the voice channel and alongside Freddy AI Agent



A ticket is created when...

- the action is unsafe or unclear
- the caller doesn't confirm
- the caller is angry / distressed
- the AI should not guess



What we add to Freshdesk

Code-mixed intent validation, action gating, transparent confirmation and structured escalation, on top of ticketing, agent workspace and customer history.

What the human agent receives in the Freshdesk ticket

CONVERSATION

Call ID + timestamp

Original code-mixed transcript

Detected languages

English summary

Voice / DTMF source

DECISION

Intent + confidence

Negation detected

Safety verdict

Requested API action

Confirmation status

OPERATIONAL

Priority / tags

SOP result

Reason for escalation

Attempted API call

Audit reference

5 Agentic AI Design & SOP Audit Trail

A plug-in control point: it intercepts the proposed action before the company API is called

1 EXISTING AI AGENT

Understands the conversation and proposes a tool / API action.

2 CODEMIX SKILL (plug-in)

Claude checks intent + negation; the gate checks risk, sentiment, SOP and confidence.

3 DECISION GATE

ALLOW → API · CONFIRM → caller · BLOCK → stop + Freshdesk

4 COMPANY SYSTEMS

Dodo / company APIs run only after the gate; Freshdesk gets the handoff.

Plugs in via an MCP / REST adapter around the agent's tool calls, whether that is Freddy AI Agent or the company's own agent. The company keeps its existing agent and APIs.

SOP-BASED AUDITING

Captured

transcript · intent · negation · confidence · sentiment · confirmation
· API action

Checked

company SOP · action authorisation · required confirmation ·
escalation policy

Produced

PASS / VIOLATION · audit reference · supervisor flag · Freshdesk
context

Every sensitive action can be reconstructed after the call:

transcript → intent → decision → confirmation → API call → SOP result
→ Freshdesk case

5

MVP Scope, What Is Built & Early Proof

What's demonstrated today, what the MVP adds, and what we've measured so far

BUILT TODAY

- Code-mix tagging engine + intent scoring
- MCP server
- Live Freshdesk ticket API
- Negation gate v0 + benchmark
- Live demo on Vercel

MVP (3 months)

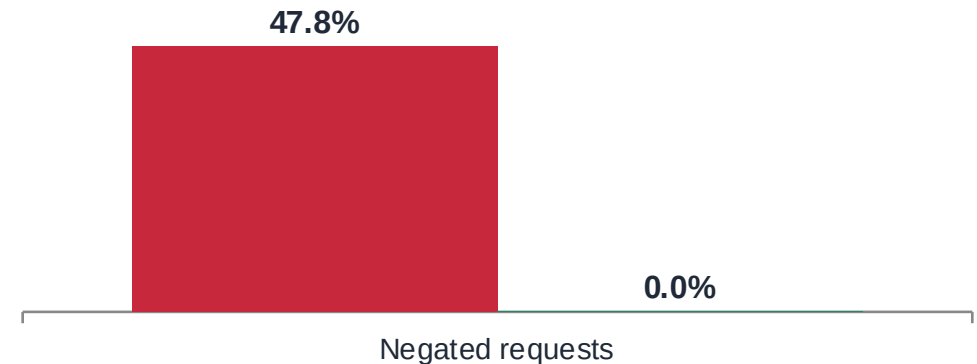
- VoBiz + Sarvam + ElevenLabs voice path
- Claude safety gate
- Dodo action path
- Freshdesk escalation
- Hindi + Tamil + English, audit trail

NEXT

- Sentiment + interrupt tuning
- SOP rule editor + reports
- More Indian languages
- Freshworks Marketplace app
- Plug-in via MCP / REST

EARLY PROOF (MEASURED)

Wrong refund / cancel on negated requests



■ Round-1 engine alone ■ With negation gate v0

11 of 23 → 0 of 23 wrong actions, with 0 of 20 genuine requests wrongly blocked.

43 team-written sentences (Hindi, Tamil, English, Telugu, Kannada, Bengali). A small, self-authored set: the pilot measures real calls. Reproduce: node decision-policy/negation-benchmark.mjs

6.7

Target Customers & Value Proposition

Start where AI agents already have permission to take customer actions



Primary

D2C, e-commerce and fintech brands using Freshdesk and deploying AI voice agents for Indian customers.



Secondary

BPOs and support outsourcers that need one safety / policy layer across many client accounts.



Protected actions

Refunds • returns • replacements • cancellations • any company API exposed to the AI agent.

Why they buy: with ~26% of COD orders returning to origin, refund / return / cancel calls are a large share of support volume, and each wrong action costs real money.

Plug it in once: the existing AI agent serves mixed-language callers, and nothing irreversible happens on a misheard word.



Without

- Acts on its first interpretation
- A dropped “mat” = a wrong refund
- Blind escalation to humans
- No record of what happened



With Codemix Skill

- Validates intent + negation first
- Confirms when unsure (voice / DTMF)
- Freshdesk handoff with full context
- Full, SOP-checked audit trail

8-9 Pricing & Packaging · Go-to-Market Plan

Pilot → measurable proof → Freshworks distribution

Free

Monitor mode

Flags risky actions, never blocks · weekly wrong-action report

\$5

per 1,000 guarded actions

Full safety gate, DTMF confirm, Freshdesk escalation, audit trail · 30 days free

Custom

Enterprise

Custom SOP rule packs · transcript retention + India data residency · SLA

Why the price works

Only sensitive actions are metered. One prevented wrong refund of ₹1,000 (~\$11) pays for over 2,000 guarded actions. Proposed; validated in the pilot.

GO-TO-MARKET

1

3 Indian D2C design partners

already on Freshdesk with AI voice agents

2

Free monitor-mode pilot

measure wrong actions without blocking

3

Publish wrong-action proof

case study + Hinglish / Tanglish demo

4

Freshworks Marketplace listing

one-click install for Freshdesk admins

5

Scale via Freshdesk + BPO partners

sold alongside Freshdesk deals, India / SEA

Why Freshworks: Freshdesk turns an unsafe AI decision into an actionable human case.

10

Costs & Resources Needed

~\$90K proposed budget for a 3-month MVP (estimates)

BUDGET BREAKDOWN (ESTIMATE)

3 engineers × 3 months

voice path, safety gate, Freshworks app



Linguist + data labelling

~5,000 Hindi + Tamil utterances



Part-time PM

pilots + pricing



API + telephony usage

Sarvam, Claude, ElevenLabs, VoBiz, Dodo



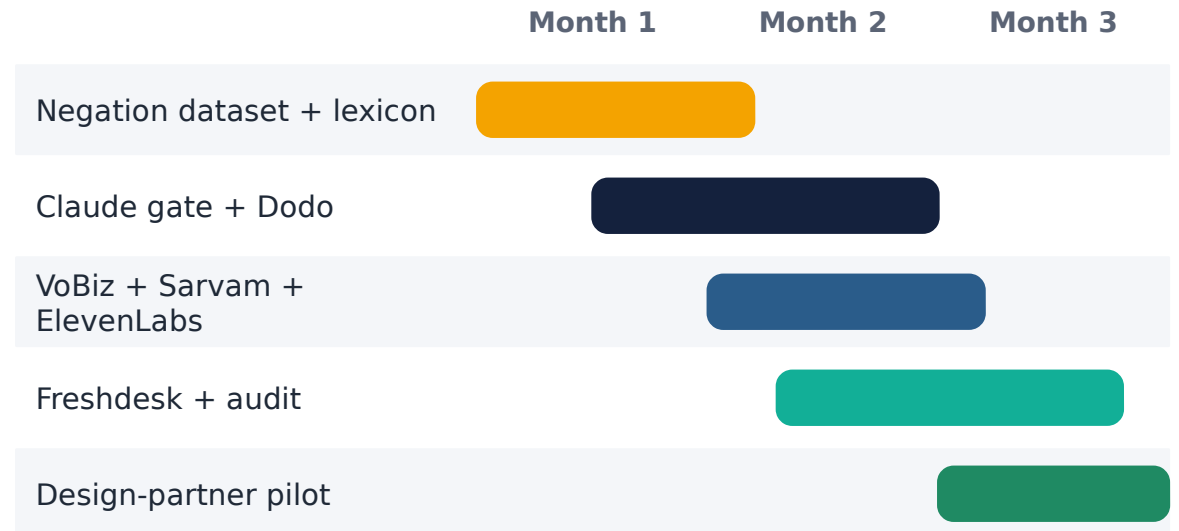
Contingency



~\$90K total

3 engineers · 1 linguist · part-time PM · 3 months

3-MONTH MVP TIMELINE



Optional revenue estimate (assumption-based)

100 customers × 20,000 guarded actions / month × \$5 per 1,000
 ≈ **\$120K ARR in Year 1**. Every input is tested in the pilot.

11

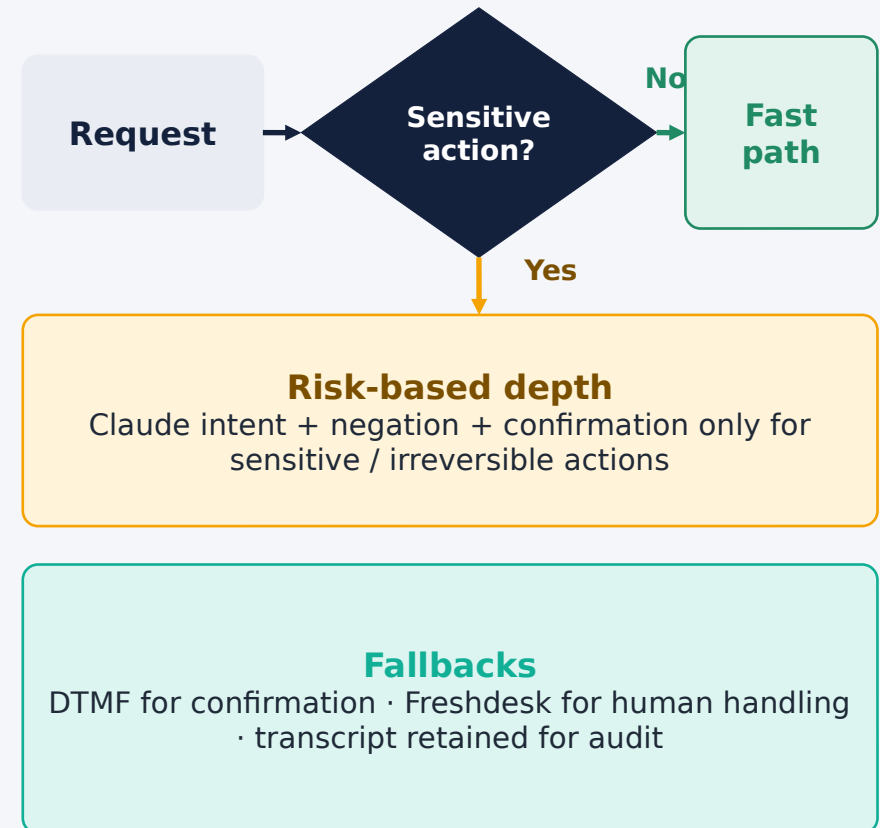
Expected Benefits & Success Metrics

Baseline where we have one; everything else is a pilot target

METRIC	TODAY	PILOT TARGET
Wrong actions on negated requests	48%*	< 0.5%
Engine latency (language + intent)	1.87 ms†	< 5 ms
p95 reply time (sensitive path)	—	< 900 ms
Confirmation rate (friction budget)	—	≤ 15%
Audit coverage	—	100%
Refund / cancel reversal tickets	—	−30%
Adoption (eligible customers, 2 quarters)	—	15%

* Round-1 engine without the gate, 23-sentence negation test set. † Average over the repo benchmark (npm test), 0.12 ms median.

HOW LOW LATENCY IS KEPT

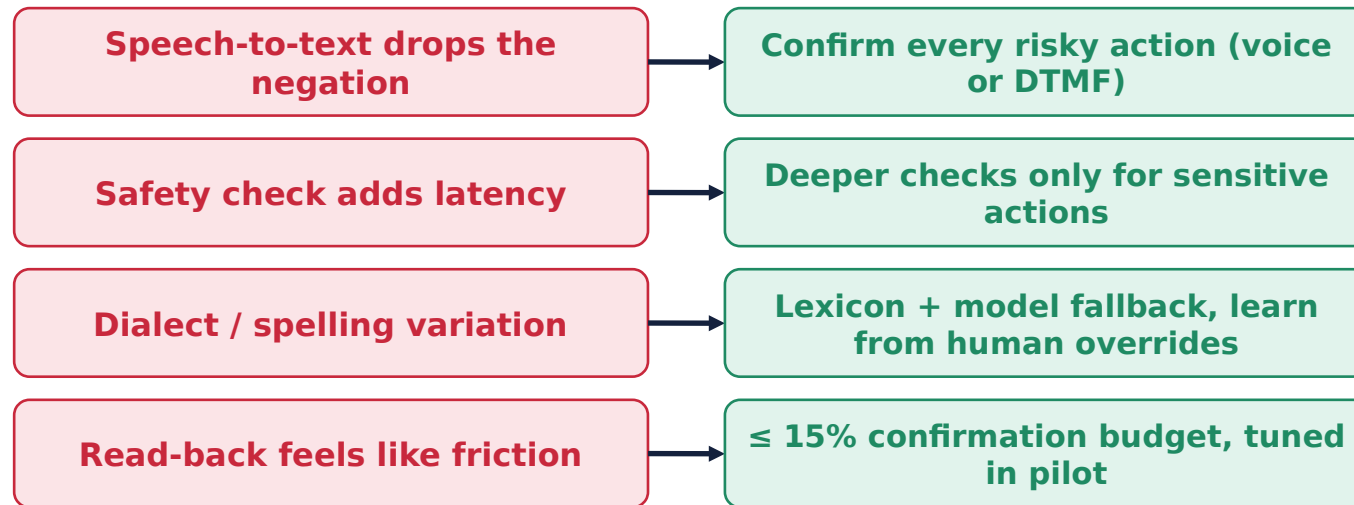


12

Risks & Assumptions

What could go wrong, how we reduce it, and what we're assuming

KEY RISKS → MITIGATION



ASSUMPTIONS

- Pilot brands expose refund / return APIs to their AI agent
- Callers accept a short confirmation on risky actions
- Sarvam accuracy on phone audio is good enough with the confirmation fallback

OUR ASK

Freshdesk sandbox + Marketplace listing path for a 3-brand pilot.

Use the pilot to measure wrong-action reduction, confirmation rate, latency and adoption.

Decision principle

Do not silently guess on a sensitive action; expose uncertainty, confirm or route to a human.

★ Final Pitch — Why This Is a Freshworks Extension

A concise closing story for the judges

● THE PROBLEM

Code-mixed voice + negation can cause an AI agent to misunderstand an action.

● THE PRODUCT

An intent-validation safety gate between what the voice agent hears and what it is allowed to do:
Understand → Decide → Act → Audit.

● THE FRESHWORKS VALUE

When the AI should not act, Freshdesk turns uncertainty into a structured human workflow with transcript + decision context.

THE DEMO



“Don't cancel” should mean don't cancel — and Freshworks should know exactly what happened when the AI chooses not to act.

Codemix Skill • Understand → Decide → Act → Audit • Thank you